

AFRI Terminal - Complete Build Specification

A Godel-Terminal-style financial research terminal for African markets (Kenya, Nigeria, South Africa, Egypt). This document is the complete specification: what it is, every feature, every data source, the architecture, the UI design, the command set, the file layout, and the build phases. Claude should be able to rebuild or continue this project from this document alone. Last updated August 2026.

1. Vision And Product Definition

1.1 What It Is

A browser-based financial research terminal modeled on Godel Terminal (godelterminal.com) and Bloomberg, localized for four African exchanges: NSE Nairobi (Kenya), NGX Lagos (Nigeria), JSE Johannesburg (South Africa), and EGX Cairo (Egypt). It is a research terminal, NOT a broker: no order execution, no portfolio management. Target users are African retail investors, analysts, and finance professionals who cannot afford Bloomberg (~\$32K/yr) or even Godel (~\$996/yr).

1.2 Core Principles

1. Bloomberg-style command UX: backtick command palette, TICKER COMMAND syntax. 2. Black background with mint green accents (Godel palette, exact colors in section 7). 3. Free or cheap data only (the builder is OUT OF MONEY - no paid data subscriptions). 4. Live data must always display, even when markets are closed (fall back to last close). 5. Full listings: every traded stock on all four exchanges. 6. Research depth: dividends, filings, financial statements, news, FX, heatmap, charts.

2. Markets And Tickers

2.1 Exchange Coverage

Country	Stocks	Live?	Ticker Format
South Africa	319	YES (Yahoo .JO)	SOL.JO, NPN.JO, AGL.JO
Egypt	259 (207 with real tickers)	YES (Yahoo .CA)	COMI.CA, CCAP.CA (short) or EGS60121C018.CA (ISIN fallback)
Nigeria	132 (101 with tickers)	EOD only (Yahoo dropped .NG)	MTNN, DANGCEM (display only)
Kenya	58 (39 with tickers)	EOD only (Yahoo dropped .NR)	SCOM, EQTY, KCB (display only)

2.2 Ticker Sources And Rules

JSE tickers come from Twelve Data's keyless stocks list API (exchange=XJSE) and work directly on Yahoo as SYMBOL.JO. EGX tickers were extracted from TradingView's screener (Python package tradingview-screener, market=egypt, 290 securities) and matched to company names by fuzzy matching (difflib, threshold 0.62); each ticker was then verified against Yahoo chart API (200 + price). 217/290 verified. Companies that fail verification keep their ISIN code as the working symbol (EGS60121C018.CA still fetches on Yahoo). NGX/NSE tickers (MTNN, SCOM...) come from TradingView screener (markets nigeria/kenya) and are DISPLAY ONLY - the data for those two exchanges is EOD from African Financials, matched by company name, because Yahoo removed .NG and .NR coverage.

IMPORTANT for the builder: the master file stocks.json stores, per stock: sym (working Yahoo symbol), code (ISIN for EGX), name, currency, and optional short/ticker (display ticker). The frontend displays short||ticker||sym-stripped, in that priority order. Heatmap labels use the same priority. The command bar accepts short tickers (COMI DES works) as well as full syms.

3. Full Feature Set

3.1 Godel Command Set (localized)

Panel	Godel Function	AFRI Localization
COMPANY	Security overview (business, price, mcap, EPS, ratings)	Company profile + quote for all 4 exchanges
NEWS	Company news	Aggregated African RSS (5 feeds)
WATCHLIST	Quote monitor / watchlists	Full listing per exchange, search + sector filter
SEC	SEC filings (EDGAR)	AF filings/documents per company (NGX/NSE); SENS-equivalent links
FINANCIALS	Standardized financials	Financial summary from latest filing (revenue, EBIT, PAT, EPS, assets)
INDICES	World indices	NSE20, NASI, NGX ASI, JSE ALSI, EGX30
EPS / EM	Earnings matrix	EPS + latest earnings from filing summary
OPTIONS	Options chains	JSE only (other markets too thin) - not yet built
SCREENING	Equity screening	SCREEN command - planned (see section 12)
CHART	Charting	TradingView lightweight-charts, 1M/3M/6M/1Y/2Y/5Y
TIME & SALES	Time & sales	Not yet built
HOLDERS	Holders	Not yet built
COMMODITIES	Global commodities	Planned: gold, platinum, cocoa, coffee, tea
FOREX	Forex	6 live pairs: USD/ZAR, USD/EGP, USD/KES, USD/NGN, EUR/ZAR, G
HELP	Command help	Toast with command list

3.2 Custom Commands And Panels

DIV, ERN, EM, FIN all map to the FUND panel. REG (regulatory wire) and BND (bonds/T-bills) are planned. The command palette is opened with the backtick key. Syntax: TICKER COMMAND (e.g. SOL DES, COMI HP, SCOM C). Bare commands: WEI, FX, NEWS, HELP, QM, FUND. Unknown commands show a toast listing valid commands.

4. UI Layout (3-Panel Bloomberg Style)

The screen is a full-viewport dark terminal with three columns plus top and bottom bars. All panels are always visible; the right panel has tabs.

, 4 exchange tabs (JSE/EGX/NGX/NSE with stock counts + live status dot), search box, live market clock per exchange timezone, market c
30 (^EGX30.CA), NSE20, NASI, NGX ASI - LIVE where available, with change % and LIVE tag
(ticker or company name), sector filter chips, rows showing display ticker + company name + last price + change %; click a row to load that
MAP toggle button), the TradingView lightweight-charts canvas (area chart + volume histogram), heatmap container (hidden by default)
adlines. FX = 6 live currency pairs. FUND = company fundamentals (profile, highlights, financial statements, dividends table, filings table).
e (LIVE / EOD), last update

Data flow: page loads -> 4 exchange listings fetched from /api/listing -> active exchange watchlist rendered immediately (names only) -> live quotes fetched for first 20 symbols (progressive fill, capped for Yahoo rate limits) -> security detail + chart loaded for first stock -> 60s refresh cycle. Every price fetch falls back to last close so closed markets still show data.

5. Heatmap (TradingView Style)

The heatmap is the flagship visual feature. Accessed via the HEATMAP button in the chart tools row (it has its own onclick handler - the range buttons loop must skip elements without data-r attribute, otherwise it overwrites the heatmap button handler).

Rendering rules: 1. Cells grouped by sector with sticky sector headers showing sector name, avg change %, company count, up/down split. 2. Cell size proportional to trading volume on a log scale (min ~46px, max ~120px), height = 42% of width. 3. Cell color from a red/green gradient keyed to % change (green = gains, red = losses, dark gray = flat/no data; the mint green theme color is used for positive). 4. Each cell shows: company logo (Google favicon via company_domains.json, 14px, white rounded background, hidden on error), the display ticker (short > ticker > sym-stripped), and the % change. 5. Hover tooltip shows full company name + change. 6. Click a cell to load that security in the main panel. 7. A legend (strong down / flat / strong up) sits at the bottom. 8. Switching exchange tabs reloads the heatmap for that market.

Data: the server /api/heatmap?exchange=X endpoint returns {sym, code, name, short, price, chgPct, volume, sector, currency, logo, date}. For JSE/EGX it fetches live quotes (chunked, cached 240s) and classifies sector by company-name keyword rules (SECTOR_RULES list in afri_server.py, ~50 keyword groups: banks, mining, oil & gas, telecom, insurance, REIT/property, retail, food & beverage, technology, industrials, healthcare, media, transport, construction, agriculture, tobacco, etc.). For NGX/NSE it uses the sector field already in stocks.json from African Financials, with keyword classification as fallback. The heatmap payload is cached 90s.

6. Fundamentals (FUND Panel)

The FUND panel (tab, or commands FUND/DIV/CF/FA/ERN/EM/FIN) shows everything known about a company. Data comes from two places: (a) African Financials company pages (NGX/NSE only - AF does not cover JSE/EGX), stored in fundamentals.json; (b) Yahoo dividend events (JSE/EGX), fetched live.

6.1 FUND Panel Sections (in order)

1. COMPANY: profile fields (founded, listed, year_end, industry, indices, transfer_secretary, phone). 2. FINANCIAL HIGHLIGHTS: labeled figures from the AF company page profile text. 3. FINANCIAL STATEMENTS (latest): revenue, operating profit/EBIT, profit after tax, EPS, total assets, DPS parsed from the meta description of the company's latest filing page on AF (both bullet format '- Label: value' and prose format 'Revenue grew to NGN 2.51 trillion' are handled; parsed by fill_financial_summary.py into the financial_summary field). 4. DIVIDENDS: full history table (date, amount, currency). For NGX/NSE this comes from AF dividend tables (dates array: post/declared/record/ex-div/payable + amount); for JSE/EGX from Yahoo chart API events=div (date + amount). 5. FILINGS & REPORTS: table of all documents (type: Interim Report/Annual Report/Presentation, year, period, date, URL link - clicking opens the PDF). 6. PER-SHARE METRICS: EPS, last dividend per share. Footer: data source + 'not investment advice'.

6.2 Data Files

fundamentals.json: keyed by 'EXCHANGE:slug' (e.g. NSE:ke-scom, NGX:ng-dangce). Each entry: {slug, name, profile{}, highlights{}, financial_summary{}, documents[{}], dividends[{}], fetched}. 190 companies (132 NGX + 58 NSE), 0 errors. Dividends: ~132 companies have them. Filings: 189 companies. Profiles: all 190. Highlights: 28 (regex-limited). financial_summary: filled by the throttled filler script (rate-limit safe, 1.2s between fetches). The server loads this file at startup; the /api/fundamentals?symbol=X&exchange=Y&name=Z endpoint returns {exchange, af, dividends} where af is the AF record and dividends are Yahoo events for JSE/EGX.

7. Visual Theme (Godel Colors)

Exact palette pulled from Godel Terminal's own CSS. Use these EXACT values:

Variable	Value	Use
--bg	#000000	Page background
--bg2	#0a0a0a	Panel background
--bg3	#141414	Sub-panel / headers
--bg4	#1f1f1f	Command bar / active
--line	#1a1a1a	Borders
--line2	#262626	Stronger borders
--tx	#f0f0f0	Primary text
--tx2	#9a9a9a	Secondary text
--tx3	#6a6a6a	Muted text / labels
--up	#33e29a	Gains (mint green - Godel signature)
--down	#f87171	Losses (red)
--acc	#33e29a	Accent / active tabs
--acc2	#1ecfb0	Secondary accent (teal)
--gold	#33e29a	Brand accent (header/legend)
--gold-dim	#0f5c3e	Muted accent

7.1 Fonts (Godel's Actual Fonts)

Google Fonts link: [family=Oxygen+Mono&family=Inter:wght@400;500;600;700;800&family=Sofia+Sans+Semi+Condensed:wght@400;600;700;800](https://fonts.google.com/specimen/Oxygen-Mono?family=Oxygen+Mono&family=Inter:wght@400;500;600;700;800&family=Sofia+Sans+Semi+Condensed:wght@400;600;700;800). Oxygen Mono for ALL data/mono text (watchlist, tables, numbers, tickers). Inter for UI text. Sofia Sans Semi Condensed for the logo/brand. Do NOT use JetBrains Mono (legacy, replaced).

7.2 Layout Details

Dark scanline-free terminal aesthetic. Font sizes: tickers 10-11px, prices 12px bold mono, headers 10px uppercase letter-spaced, section heads 10px 800-weight with mint color. Rows: hover background --bg3, bottom border rgba(38,38,38,.5). Scrollbars styled dark. The toast notification slides in at top center. Loading spinner: thin mint ring.

8. Data Sources (All Verified Working)

	Auth
ice, prev close, day high/low, volume, 52W, currency, marketState), dividend events (events=div)	None
s with short symbols	None for lists
L tickers + sectors for egypt (290), nigeria (137), kenya (57)	None
file, dividends, documents/filings, share-price tables (EOD), financial summary in filing meta descriptions	None (browser UA + Accept-Lan
eatmap	None
	None
	None

KNOWN BLOCKED (do not waste time): Mubasher API (403), EGX website (TSPD bot challenge), investing.com (403), Yahoo quoteSummary/fundamentals endpoints (401 - dead), StockAnalysis (no JSE/EGX coverage), Wikipedia EGX company tables (don't exist). JSE/EGX income statements have NO free structured source - the FUND panel for those exchanges shows dividends (live) + quote data, with an honest note that full statements need licensed feeds (Bloomberg/Refinitiv/IRESS).

9. Server Architecture (afri_server.py)

Python 3 stdlib ONLY (http.server, urllib, json, threading, concurrent.futures). No pip dependencies. Run: python afri_server.py -> serves on http://127.0.0.1:8081/. Serves terminal.html + static files, and JSON APIs. Cache: in-memory dict {key: (expires, payload)} with per-type TTLs (chart 45s, quote 240s, fx 60s, news 300s, dividends 86400s, heatmap 90s). A background warm thread fetches all JSE+EGX symbols at startup so first page load is instant (the full 578-symbol warm takes ~3 min under Yahoo throttling - run it, cache, then serve).

9.1 API Endpoints

	Returns
	ok
ange=JSE EGX NGX NSE	full listing {exchange, count, stocks[]} with sym/code/name/short/ticker/currency/price (E
ols=A,B,C	live quotes per symbol (parallel, chunked 20, throttle-safe)
=X&range=1m 3m 6m 1y 2y 5y&interval=1d 1wk	OHLC bars + meta (via Yahoo, cached)
hange=X	{exchange, count, stocks[]} with price/chgPct/volume/sector/short/logo
?symbol=X&exchange=Y&name=Z	{exchange, af, dividends} - AF record + Yahoo dividends for JSE/EGX
	6 live currency pairs
	aggregated RSS headlines (40+ items)
e=X&name=Y	EOD quote from stocks.json

9.2 Rate-Limit Handling (Critical)

Yahoo throttles bursts from datacenter IPs: single calls take ~4.7s. Strategy: (1) quote_many fetches in chunks of 20 with ThreadPoolExecutor(8) and 0.8s sleep between chunks; (2) aggressive caching with the TTLs above so each symbol is fetched at most once per 4 minutes; (3) startup warm thread pre-fetches everything; (4) the frontend caps first-pass quote fetches at 20 symbols (progressive fill, never blocks UI); (5) heatmap uses the same cache so after one build it's instant. African Financials: max 2 concurrent workers, 0.6s sleep in the scraper, 1.2s in the financial-summary filler.

10. File Layout

	Purpose
	Single-file frontend: all CSS + JS inline, TradingView lightweight-charts via CDN, Google Fonts
	Local proxy server: Yahoo proxy, RSS news, listings, heatmap, fundamentals, caching, warm thread
	Master dataset: 768 stocks across JSE(319)/EGX(259)/NGX(132)/NSE(58) with sym/code/name/ticker/currency
on	AF company data: profiles, dividends, filings, highlights, financial_summary (190 companies)
ns.json	172 curated company-name-fragment -> domain mappings for Google favicon logos
ers.json	217 EGX tickers verified against Yahoo
nigeria.csv, tv_kenya.csv	TradingView screener dumps (ticker, description, sector)
	AF company slug map (ex -> [{slug, name}])
	Rebuilds stocks.json from Twelve Data + AF tables
als.py	Scrapes AF company pages: profile/dividends/documents/highlights (resumable, 190 companies)
mary.py	Throttled pass to parse financial_summary from latest filing meta descriptions
FINAL-BUILD-SPEC.md	Original build spec (Godel feature map, data matrix, phases)

11. Build Phases

Item	Status
Foundation: Yahoo adapter (JSE+EGX live), Twelve Data listings, AF EOD scrapers, normalized quotes table	DONE
Initial shell: 3-panel layout, exchange tabs, watchlist, live clock, status line	DONE
Charts: TradingView lightweight-charts, ranges, volume	DONE
Regulatory wire (5 RSS), FX panel (6 pairs), index strip	DONE
Heatmap: sectors, volume sizing, colors, logos, tickers	DONE
Fundamentals: AF scrape (dividends/filings/profiles), Yahoo dividends for JSE/EGX, FUND panel	DONE
Financial statements: financial_summary parser from filing meta descriptions	IN PROGRESS (fill_financials)
Equity screener (SCREEN command): filter 768 stocks by exchange/sector/price/%chg/dividend yield	PLANNED - see section 12
Regulatory command (regulatory wire): CMA/SEC/FSCA/FRA/central-bank releases	PLANNED
T-bills command: T-bills and bonds board per country	PLANNED
Deployment: free hosting (Cloudflare Pages / Railway), domain, auth	PLANNED

11.1 Commands To Implement Next

SCREEN (equity screener): UI tab or command with filters - exchange multiselect, sector dropdown, min/max price, min/max % change, dividend-yield min, sortable results table. Data source: heatmap payload already has price/chgPct/sector for all stocks; dividends come from fundamentals.json (NGX/NSE) and Yahoo events (JSE/EGX); yield = last dividend / price. REG (regulatory wire): aggregate RSS/announcement feeds from CMA Kenya (cma.or.ke), SEC Nigeria (sec.gov.ng), FSCA SA (fsca.co.za), FRA Egypt (fra.gov.eg), central banks (CBK, CBN, SARB, CBE) into a scrolling feed; all sites verified reachable except Nigeria DMO (403). BND: T-bill auction results per country (CBK, DMO, SARB, MoF).

12. Known Issues And Honest Limits

1. JSE/EGX full income statements: no free structured source exists (verified: Yahoo fundamentals 401, Twelve Data keyed, StockAnalysis no coverage, AF no coverage). FUND shows dividends + quote data for those. 2. NGX/NSE live quotes: Yahoo removed .NG/.NR coverage; only EOD from AF (updated daily). 3. EGX: 52 of 259 companies still use ISIN codes as symbols (no short ticker found in any free source); they still fetch. 4. Yahoo rate limits from datacenter IP: mitigated by caching/chunking/warm-thread; on a residential IP it is much faster. 5. Sector classification for JSE/EGX is keyword-based (~15% fall into 'Other') - acceptable. 6. Heatmap logos resolve for ~50% of companies (Google favicon); the rest show a clean initial-letter avatar - acceptable. 7. fundamentals.json highlights regex catches ~28/190 (profile text varies); financial_summary from filing meta descriptions is more reliable. 8. The server dies when the Hermes background process is cleaned up between sessions - restart with python afri_server.py; a start.bat launcher is recommended. 9. Command bar: FUND/DIV/CF/FA/ERN/EM/FIN all open the FUND panel; OMON/TAS/HMS/TRAN not built.

VERIFIED DEAD ENDS (do not retry): Mubasher API (403 even with Referer), EGX site (TSPD challenge), investing.com (403), Yahoo quoteSummary & fundamentals-timeseries (401/400), Wikipedia EGX lists (no table exists), Zawya market pages (404).

13. How To Run

1. cd D:\heremes mutxri && python afri_server.py. 2. Open <http://127.0.0.1:8081/terminal.html> in Chrome/Edge. 3. First load: warm thread pre-fetches JSE+EGX symbols (~3 min under datacenter throttling; page is usable immediately, prices fill progressively). 4. Refresh fundamentals: python build_fundamentals.py (full rescrape) or python fill_financial_summary.py (financial_summary pass only). 5. Rebuild listings: python build_stocks.py. 6. Server binds 127.0.0.1:8081; LAN access requires binding 0.0.0.0 (then <http://:8081/terminal.html>).

END OF SPECIFICATION. This document is the single source of truth for rebuilding or continuing the AFRI Terminal. All data sources were verified working on the date above.